ANDHRA PRADESH CAPITAL REGION DEVELOPMENT AUTHORITY (AP CRDA)
AMARAVATI LEGISLATIVE ASSEMBLY COMPLEX · TENDER #996686 (REF: NIT No. 04/EE/CRDA/MEP/2026-27)

ELECTRICAL SCHEMATIC NOTES & DESIGN PARAMETERS:
1. INCOMING POWER: Dual 11kV independent feeders from APTRANSCO 132/33/11kV Substation via 3C x 300 sq.mm XLPE cables.
2. TRANSFORMERS: 2 x 2500 kVA, 11kV/433V Dyn11 cast resin dry type transformers with +/- 10% on-load tap changers (OLTC).
3. MAIN LT SWITCHBOARD: 4000A, 4-Pole, 65kA 1-sec fault-rated air circuit breakers (ACB) with microprocessor trip units (LSIG).
4. BUS COUPLER: 4000A 4-Pole bus coupler mechanically and electrically interlocked to prevent parallel operation of transformers.
5. EMERGENCY DG BACKUP: 2 x 1500 kVA silent containerized DG sets with auto-mains failure (AMF) synchronizing panel.
6. POWER FACTOR CORRECTION: 2 x 750 kVAr capacitor banks with 7% detuned reactors for harmonic suppression (THD < 5%).

Feeder Tag	Connected Load	Breaker Rating	Cable Specification	Destination
F-01 (Incomer 1)	2500 kVA Transformer 1	4000A 4P 65kA ACB	3 Runs of 4C x 300 sq.mm	Transformer TR-01
F-02 (Incomer 2)	2500 kVA Transformer 2	4000A 4P 65kA ACB	3 Runs of 4C x 300 sq.mm	Transformer TR-02
F-03 (Bus Coupler)	Main LT Bus Section	4000A 4P 65kA ACB	4000A Copper Busbar	Bus Section 1 & 2 Interlink
F-04 (Chiller 1)	450 kW (Chiller CH-01)	1000A 3P 50kA ACB	2 Runs 3.5C x 300 sq.mm	Chiller Plant Room Panel
F-05 (Chiller 2)	450 kW (Chiller CH-02)	1000A 3P 50kA ACB	2 Runs 3.5C x 300 sq.mm	Chiller Plant Room Panel
F-06 (AHU Panel)	320 kW (AHU Starters)	630A 3P 36kA MCCB	2 Runs 3.5C x 185 sq.mm	AHU Motor Control Center
F-07 (Elevators)	250 kW (Lifts 1 to 6)	500A 3P 36kA MCCB	1 Run 3.5C x 240 sq.mm	Lift Machine Room Panel

AMARAVATI GOVERNMENT COMPLEX · LEGISLATIVE ASSEMBLY		
DWG TITLE: 11KV SUBSTATION & 433V MAIN DISTRIBUTION SING		
DRAWING NO: SL-288-EL0-050-ELE-DWG-01	REV: A1 SING	SCALE: 1:100
CLIENT: AP CRDA / GOVT OF ANDHRA PRADESH	DATED: 28-09-2026	